# MILITARY SPECIFICATION

## OSCILLATOR, DOPPLER, COMPONENT BOARD ASSEMBLY: 10559694

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the Doppler Oscillator Component Board Assembly, 10559694, herein referred to as the component board assembly, used in Doppler Oscillator, 10548833. The doppler oscillator is a subassembly of the Radar Receiver R-1475/VPS-2, a part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-F-13926 | Fire Control Materiel: General Specification Governing the Manufacture and Inspection of |
| MIL-P-14232 | Parts, Equipment and Tools for Army Material; packaging and Packing of |
| MIL-I-45607 | Inspection Equipment, Acquisition, Maintenance and Disposition of |
| MIL-C-46357 | Computer, Gun Direction; M18 Manufacturing Processes and Inspection of |
| MIL-O-50665(MU) | Oscillator, Doppler: 10548833 |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Frankford Arsenal, ATTN: SARFA-MDM, Phila., Pa. 19137 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL- STD-109 | Quality Assurance Terms and Definitions |
| .MIL-STD-810 | Environmental Test Methods |

DRAWINGS

U. S. Army, Frankford Arsenal

| | |
|---|---|
| F10549694 | Component Board Assembly, Doppler Oscillator |
| D10548870 | Schematic, Doppler Oscillator |

PACKAGING DATA SHEET

| | |
|---|---|
| 10559694 | Packaging of Component Board Assembly |

(Copies of specifications, standards, drawings and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity-or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The component board assembly shall be manufactured in accordance with Drawing F10559694 and drawings pertaining thereto.

3.2 General specifications. - Where applicable, the component board assembly shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:



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3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2 MIL-C-46357.

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. - The component board assembly shall be mounted on the vibration table in normal service position. The component board assembly shall meet the requirements of 3.4 and 3.5 after being subjected to continuous vibration for 2 hours along each of three mutually perpendicular axes within the frequency range of 10 to 55 hertz (Hz) and a half amplitude excursion of 1/64 inch. The rate of change of frequency at each frequency range shall not exceed 1 Hz per second. Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. - The component board assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The half sine shock impulses shall have an acceleration of 15 gravity units (G) and a pulse duration of 11 ± 1 milliseconds (ins). The G value of each impulse shall be within ± 10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5-1/2 ms.

3.3.3 Altitude.

3.3.3.1 OPerating. - The component board assembly shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating. - The component board assembly shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).



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3.3.4 Temperature.

3.3.4.1 Operating. - The component board assembly shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +155°F.

3.3,4.2 Nonoperating (storage). - The component board assembly shall not suffer permanent damage when subjected to not less than 4 hours at each temperature extreme of -65°F and +-160°F, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3,3.5 Humidity (nonoperating). - The component board assembly shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity of 95 ± 5 percent for a period of not less than 10 days.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the runs list 10559694 of the doppler oscillator.

3.5 Performance. - Connect a lk-ohm ± 10% variable resistor to terminals E3, E4, and E5 of the component board assembly as shown in figure 1. Connect an 82-ohm ± 5% 1 watt resistor between terminals E6 and E7 of the component board assembly. With the 28 Vdc and 15 Vdc applied as specified in table I, the component board assembly shall produce the results as given in 3.5.l. and 3.5.2,

3.5.1 Output signal.

a. With variable resistor fully counterclockwise, E3 to E4, the output signal at E6, with respect to ground, shall be 550 ± 15G millivolts (rev) peak-to-peak at 1.5 ± 0.5 kilohertz (kHz).

b. With variable resistor fully clockwise, E4 to E5, the output signal at E6, with respect to ground, shall not be greater than 25 mv peak-to-peak at 1.5 ± 0.5 kHz.

c. With variable resistor set as in b. above, the dc output voltage at E6 shall be 80 ± 10 mv.

3.5.2 Output bias voltage. - With only the 15 Vdc applied to the terminals as specified in table 1, the component board assembly shall have a dc bias output of -410 ± 10 mVdc at E6 with respect to E9 (GND).



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TABLE I; - INPUT VOLTAGES FOR COMPONENT BOARD ASSEMBLY

| VOLTAGE | COMPONENT BOARD TERMINALS |
|---|---|
| +28.0 ± 0.1 Vdc/50 mA (maximum) | El (POS) |
| +28 V (Return) | E8 (RET) |
| -15.0 ± 0.1 Vdc/25 mA (maximum | E7 (NEG) |
| -15 V (Return) | E8 (RET) |

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial Production) submission. - The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three component board assemblies manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.



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4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection Provisions.

4.3.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4,3.2 Examination and tests.

4.3.2.1 Components and subassemblies. - All components and sub-assemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provision (SQAP) listed in the technical data package (TDP). In the absence-of SQAP'S, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. - Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in 4.3.2.3 and 4.4 herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.3.2.3 and 4.4 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests. - The requirements and tests in table II shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature (60°F and 90°F).



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**TABLE II. - CLASSIFICATION OF DEFECTS**
(100 percent inspection)

| CHARACTERISTIC | REQUIREMENT | TEST PROCEDURES |
|---|---|---|
| **CRITICAL: NONE** | | |
| **MAJOR:** | | |
| 101. Resistance | 3.4 | 4.6.2 |
| 102, Performance | 3.5 | 4.6.3 |
| **MINOR: NONE** | | |

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling. - One component board assembly shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in table III. Three component board assemblies shall be selected at random by a Government representative as a control sample from each SO produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in table IV.

**TABLE III. - SPECIAL SAMPLING**

| CONTROL TEST | REQUIREMENT | TEST PROCEDURES |
|---|---|---|
| 301. Fabrication | 3.1 | Applicable drawings-Visual |
| 302. General specifications | 3.2 | MIL-F-13926 & MIL-C-46357-Visual |



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TABLE IV. - SPECIAL SAMPLING, ENVIRONMENTAL .

| CHARACTERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|
| 303. Vibration | 3.3.1 | 4.6.1.1 |
| 304 . Shock | 3.3.2 | 4.6.1.2 |
| 305 . Altitude | 3.3.3 | 4.6.1.3 |
| 306. Temperature | 3.3.4 | 4.6.1.4 |
| 307 . Humidity (nonoperating) | 3.3*5 | 4.6.1.5 |

4.4.1 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections halve been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment. - Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.



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4.5.2.3 Test equipment. - In conjunction with 4.5.2.2, the standard test equipment listed in table V shall be utilized in performing the applicable tests as specified in 4.6.

TABLE V. - TEST EQUIPMENT LIST

| ITEM NUMBER | | DESCRIPTION |
|---|---|---|
| 1. | Power supply | Dc, dual output, O to 40 Vdc at 0.5A, regulated to 0.01%. Ripple 2 mV pk-pk maximum. |
| 2. | Oscilloscope | Vertical sensitivity: 10 mV/division. Vertical passband: 100 kHz. Sweep rate: 5 kHz. |
| 3. | VOM, digital | Dc, 0-50 V, ± 0.1%. To measure less than 2 ohms. |
| 4. | $'_1$ | Potentiometer, lk ± 10%, 1 watt. |
| 5. | $R_2$ | Resistor 82 ohm ± 5%, 1 watt. |

4.6 Test methods and procedures.

4.6.1 Environmental. - Environmental test shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.6.1.1 Vibration. - The test shall be performed as specified in Method 514, Procedure IX, Part I of MIL-STD-810 except for conditions in 3.3.1 and that failure criteria do not apply. After vibration, the component board assembly shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.1

4.6.1.2 Shock. - This test shall be performed as specified in Procedure I, Method 516 of MIL-STD-810 except for conditions specified in 3.3.2. After shock impulses, the component board assembly shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.2.

4.6.1.3 Altitude. (nonoperating and operating). - This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in step 2 the altitude shall be limited to 40,000 feet (5.5 inches Hg). Operation of step 3 shall be the performance tests specified in 4.6.3.



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4.6.1.4 Temperature.

4.6.1.4.1 High (operating and nonoperating). - This test shall be performed as specified in Procedure I, Method 501 of MIL-STD-810. The temperature of step 2 shall be +160°F. The temperature of step 3 shall be maintained for not less than 4 hours. The temperature of step 4 shall be +155°F. In step 5, the temperature shall be maintained for 4 hours and the component board assembly shall be tested in accordance with 4.6.2 and 4.6.3 during this time.

4.6.1.4.2 Low (operating and nonoperating). - This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of step 2 shall be -65°F maintained for not less than 4 hours. The temperature of step 4 shall be -40°F. In steps 5 and 7, the component board assembly shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.5 Humidity (nonoperating. - This test shall be performed in accordance with Procedure II, Method 507 of MIL-STD-810. The component board assembly shall be tested in accordance with 4.6.3.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance. - Connect the component board assembly as shown in figure 1 and adjust power supply, item 1, to the input voltages as specified in table I.

4.6.3.1 Output signal. - Using the oscilloscope, item 2, observe the signal condition at E6 of the component board assembly as specified in 3.5.l.a and 3.5.l.b. Using the digital voltmeter, item 3, observe the signal condition as specified in 3.5.1.c.

4.6.3.2 Output bias voltage. - Disconnect the positive 28 Vdc input lead from El of the component board assembly. Using the digital voltmeter, item 3, observe the dc bias voltage as specified in 3.5.2.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking: - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10559694. The level of protection shall be as specified in the procurement document.



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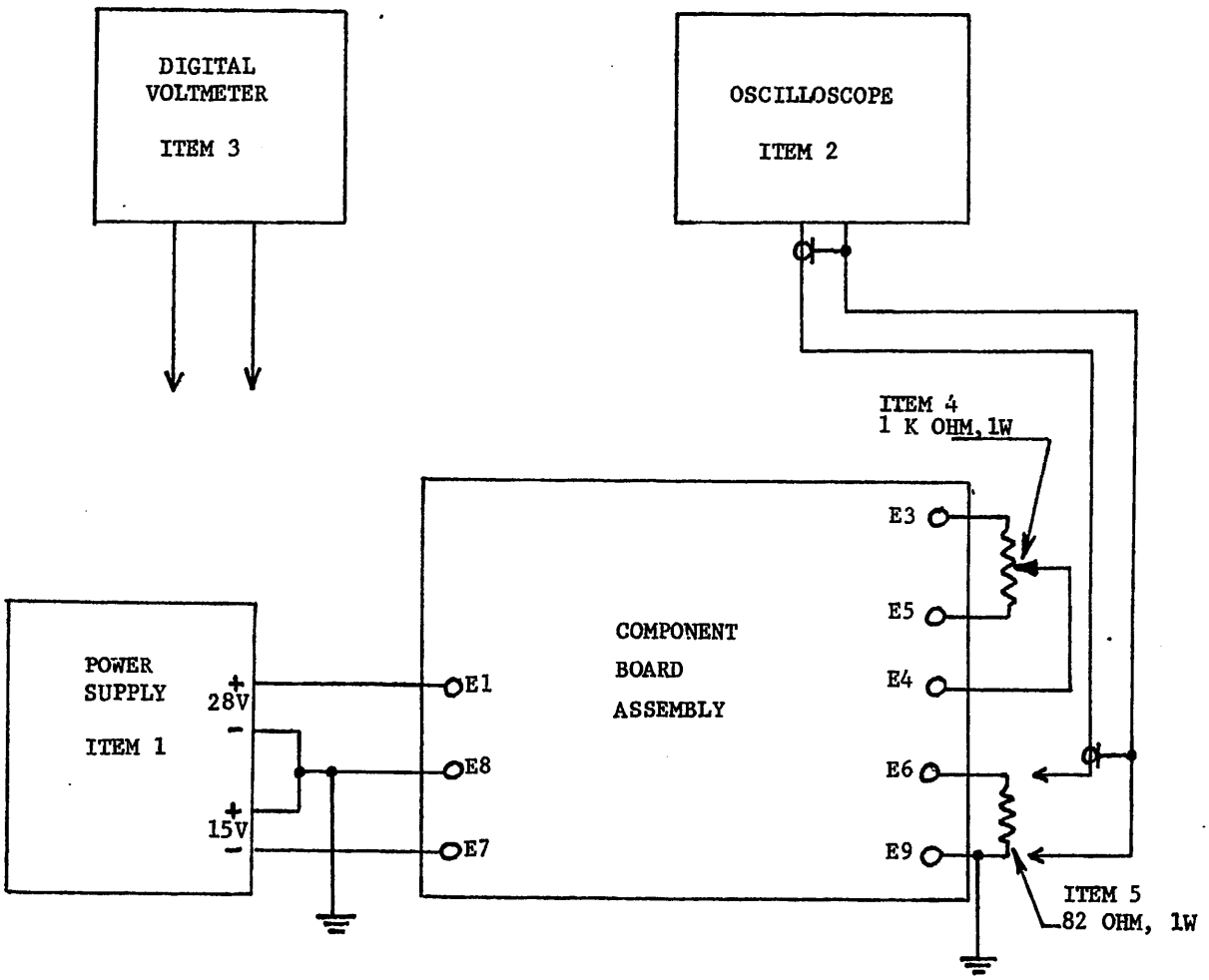



Figure 1. - Test set up



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## 6. NOTES

**6.1** Intended use. - The component board assembly is used in the doppler oscillator, a subassembly of the Receiver R-1475/VPS-2. The **doppler oscillator provides a dc bias voltage to the doppler modulator at all times and a** 1500 Hz modulation signal to produce **an** artificial **target signal during system test mode (clutter lock-on/mid-range calibration).**

6.2 Ordering data. - Procurement documents should specify the following:

**a. Title, number, and date of this specification.**

b. **Applicable packaging data sheet number (see 5.1).**

**c. selection of applicable level of preservation packaging and packing.**

d. **Level of protection required.**

**e. Applicable stock number.**

**f. Requirement for first article submission.**

**Custodian:**

**Army - MU**

**Preparing Activity:**

**Army - MU**

**Project No. 1285-A375**



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DOCUMENT IDENTIFIER AND TITLE MIL-O-48434(MU), Oscillator, doppler, Component Board Assembly: 10559694

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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